Lifted Closure-Incidence Symmetry of a Named Finite Core

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Abstract

The frozen C72 named coordinates realize a finite abelian core $Q \cong \mathbb{Z}/9 \oplus \mathbb{Z}/3 \oplus \mathbb{Z}/2$. We study the incidence relation between the sixteen named labels and the cyclic subgroups they generate. The nine nonempty closure fibers have sizes $5, 1, 2, 1, 2, 1, 1, 1, 2$. Exactly twelve automorphisms of Q preserve this weighted closure profile. Retaining the ambient automorphism and lifting by all compatible label permutations gives an exact group of order $12 \cdot 5!\,2!\,2!\,2! = 11520$. A faithful GAP representation identifies the structure as $\mathrm{S}_5 \times C_2 \times D_8 \times C_6$. The action on the twenty abstract subgroups alone has image order eighteen and is therefore not the C75 symmetry object.

1 Frozen incidence object

Let

$$Q = \mathbb{Z}/9 \oplus \mathbb{Z}/3 \oplus \mathbb{Z}/2$$

and let $(x_i)_{i=1}^{16}$ be the C72 coordinate list in the frozen basis. For a label i define its cyclic closure

$$F(i) = \langle x_i \rangle \le Q, \qquad I(i, C) = 1 \iff C = F(i).$$

The C72 list hits nine nonempty cyclic subgroups. Their fiber sizes are

$$5, 1, 2, 1, 2, 1, 1, 1, 2.$$

The first factor five comes from the zero coordinate; the final factor two comes from the distinct coordinates $(4, 2, 0)$ and $(2, 1, 0)$, which generate the same order-nine cyclic subgroup. We use the scope firewall `NO_BAD_EULER_OR_ROOT_NUMBER`.

2 Ambient automorphisms and weighted stabilizer

Every endomorphism of the odd part has the form

$$(x, y) \longmapsto (ax + 3by \bmod 9,\ cx + dy \bmod 3).$$

Direct bijectivity enumeration on all 54 points gives $|\operatorname{Aut}(Q)| = 108$. For every subgroup $C \le Q$, put

$$w(C) = \#\{i : F(i) = C\}.$$

Define

$$K = \{\alpha \in \operatorname{Aut}(Q) : w(\alpha C) = w(C) \text{ for every } C \le Q\}.$$

Theorem 1 (Weighted ambient stabilizer). *The exact enumeration gives $|K| = 12$. It is generated by*

$$r = (2, 0, 0, 2), \qquad s = (2, 1, 2, 1),$$

where $|r| = 6$, $|s| = 2$, and $rs = sr$. Thus $K \cong C_6 \times C_2$.

The producer checks the weighted profile on all twenty subgroups, not merely on the nine occupied cyclic subgroups. This prevents accidental promotion of an unweighted support coincidence to a symmetry.

3 Lifted group

A lifted symmetry is a pair (α, π) with $\alpha \in K$ and $\pi \in \mathrm{Sym}(16)$ such that

$$\langle x_{\pi(i)} \rangle = \alpha(\langle x_i \rangle) \qquad (1 \le i \le 16).$$

For a fixed α, every bijection between each source fiber and its target fiber is allowed. Hence

$$|\widetilde{G}| = |K| \prod_{C} w(C)! = 12 \cdot 5!\, 2!\, 2!\, 2!\, 2! = 11520. \tag{1}$$

Proposition 1 (Exact lifted enumeration)**.** *The producer directly enumerates all* 11520 *pairs and independently obtains the same set by closure under six explicit generators: four label-fiber permutations and canonical lifts of r and s. The projection $\widetilde{G} \to K$ has kernel of order* 960*, with candidate kernel*

$$S_5 \times C_2 \times C_2 \times C_2.$$

The exact order distribution is

order	1	2	3	4	5	6	10	12	15	20	30	60
count	1	623	62	1168	24	4066	552	3296	48	192	1104	384.

GAP receives the faithful permutation action on 54 ambient points together with 16 labels and returns

$$\widetilde{G} \cong C_2 \times C_6 \times S_5 \times D_8,$$

which is the same candidate structure as $S_5 \times C_2 \times D_8 \times C_6$.

4 The non-faithful lattice boundary

All 108 automorphisms act on the complete twenty-subgroup lattice. That action has image order 18 and kernel order 6. It forgets both the ambient-map distinctions in the kernel and all label-fiber permutations. In contrast, the C75 lifted projection has kernel order 960 and retains the ambient automorphism in each pair. Therefore the eighteen-element lattice image is a diagnostic quotient, not the lifted closure-incidence symmetry.

5 Reproducibility and scope

The source-bound producer, an independent checker, a GAP cross-check, clean replay, and hostile semantic mutations all pass. The evidence records the complete twenty-subgroup list, all nine occupied closure fibers, the twelve weighted stabilizer matrices, all exact counts, and the non-faithful lattice boundary. No full table of marks, full Burnside ring, arithmetic/local construction, Euler factor, root number, automorphy, or Hilbert–Polya claim is made.